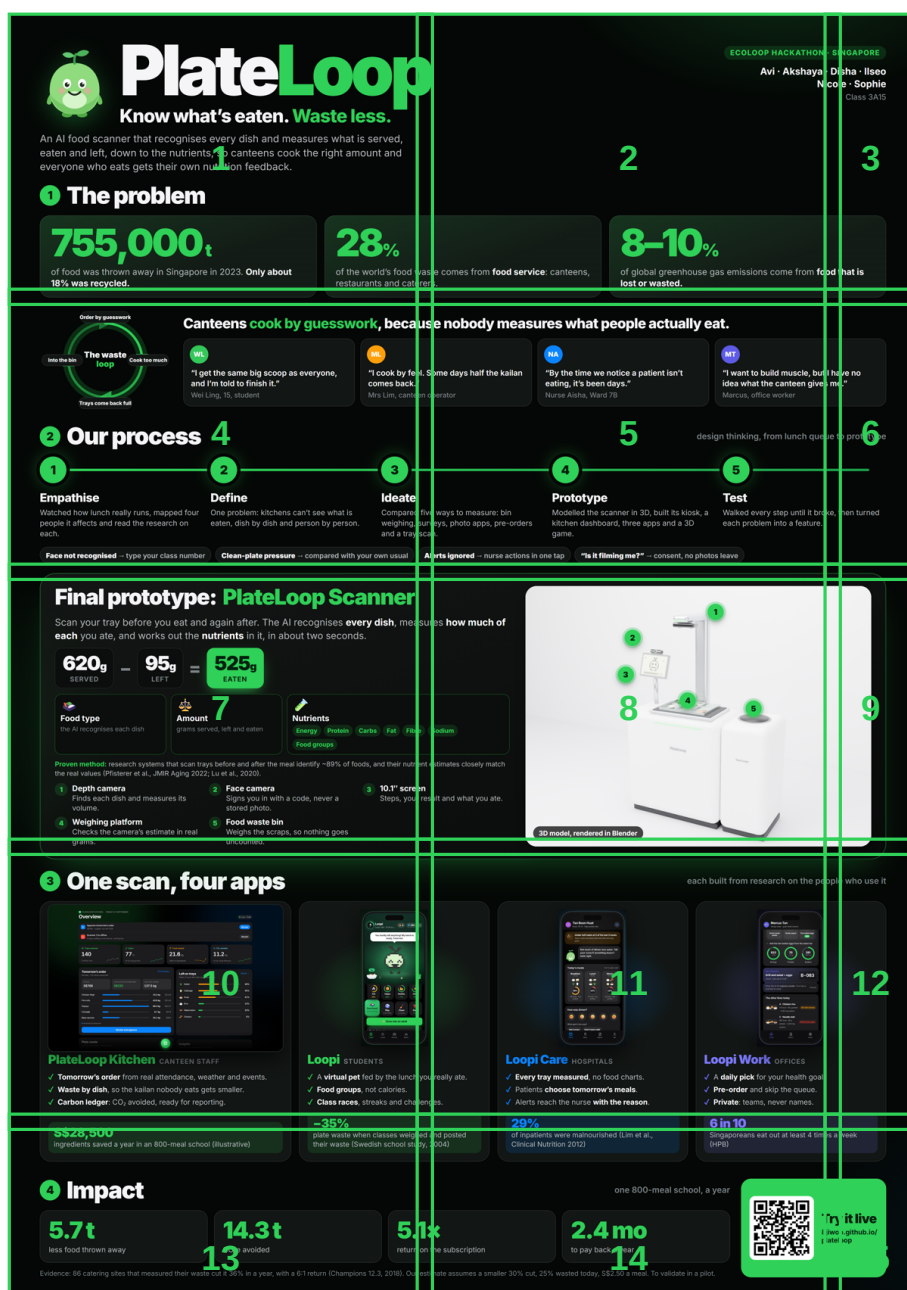


1. Lay the sheets out as numbered below: 3 across, 5 down.
2. On every sheet except the left column, cut along the LEFT crop marks.
3. On every sheet except the top row, cut along the TOP crop marks.
4. Slide each cut edge over its neighbour's 10 mm overlap until the picture lines up, then tape it from the back.
5. Trim the white margins around the outside last. The finished poster is exactly 594 x 841 mm.





PlateLoop

Know what's eaten. Waste less.

An AI food scanner that recognises every dish and measures what is served, eaten and left, down to the nutrients, so canteens cook the right amount and everyone who eats gets their own nutrition feedback.

1 The problem

755,000_t

of food was thrown away in Singapore in 2023. **Only about 18% was recycled.**

28%

of the world's food waste
restaurants and caterers

ECOLOOP HACKATHON · SI

Avi · Akshaya · Di
Nicol

waste comes from **food service**: canteens,
erers.

8–10%

of global greenhouse gas emissions come from **food t**
lost or wasted.

• **SINGAPORE**

• **Disha** • **Ilseo**
nicole • **Sophie**

Class 3A15

od that is



Canteens **cook by guesswork**, because

WL

"I get the same big scoop as everyone, and I'm told to finish it."

Wei Ling, 15, student

ML

"I cook by feel. So much food comes back."

Mrs Lim, canteen owner

2

Our process

1

Empathise

Watched how lunch really runs, mapped four people it affects and read the research on each.

2

Define

One problem: kitchens can't see what is eaten, dish by dish and person by person.

3

Ideate

Compared five different weighing, surveying and a tray scanner.

Face not recognised → type your class number

Clean-plate pressure → compared with your own usual

All

use nobody measures what people actually eat.

el. Some days half the kailan
"
en operator

NA

"By the time we notice a patient isn't eating, it's been days."
Nurse Aisha, Ward 7B

MT

"I want to build muscle, but I ha
idea what the canteen gives me
Marcus, office worker

design thinking, from lunch queue to



Prototype

Modelled the scanner in 3D, built its kiosk, a kitchen dashboard, three apps and a 3D game.

d five ways to measure: bin
surveys, photo apps, pre-orders
scan.

Alerts ignored → nurse actions in one tap



Test

Walked every step until it broke, t
each problem into a feature.

"Is it filming me?" → consent, no photos leave

I have no
s me."

ue to prototype

ke, then turned

Final prototype: **PlateLoop Scanner**

Scan your tray before you eat and again after. The AI recognises **every dish**, measures **each** you ate, and works out the **nutrients** in it, in about two seconds.

620g

SERVED

—

95g

LEFT

=

525g

EATEN



Food type

the AI recognises each dish



Amount

grams served, left and eaten



Nutrients

Energy

Protein

Carbs

Fat

Fibre

Food groups

Proven method: research systems that scan trays before and after the meal identify ~89% of foods, and their nutrient values are close to the real values (Pfisterer et al., JMIR Aging 2022; Lu et al., 2020).

1

Depth camera

Finds each dish and measures its volume.

2

Face camera

Signs you in with a code, never a stored photo.

3

10.1" screen

Steps, your results

4

Weighing platform

Checks the camera's estimate in real grams.

5

Food waste bin

Weighs the scraps, so nothing goes uncounted.

asures **how much of**

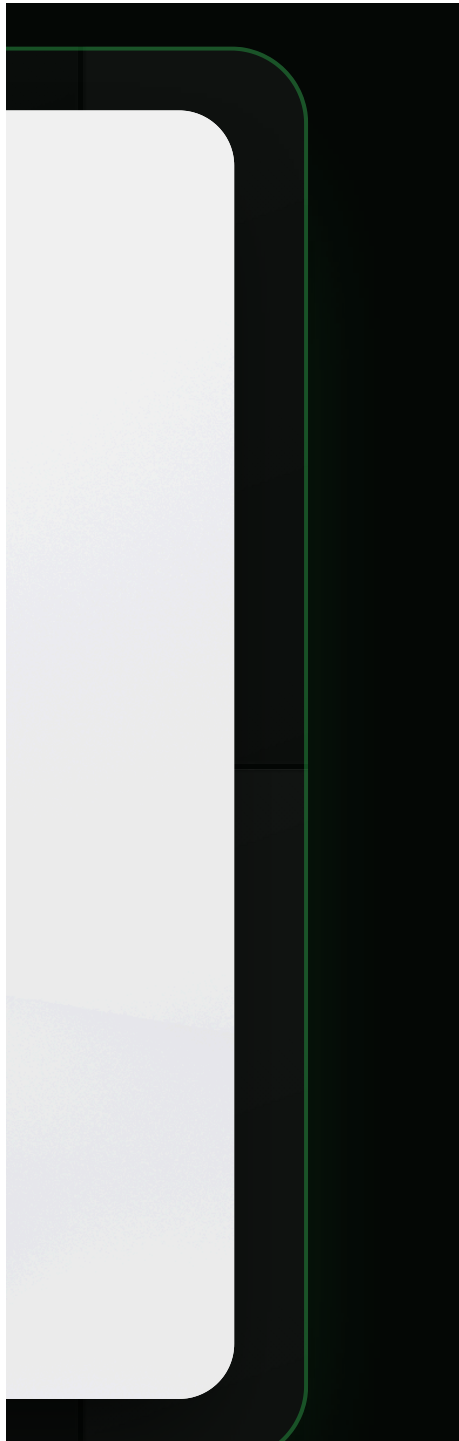
re **Sodium**

ent estimates closely match

en
r result and what you ate.



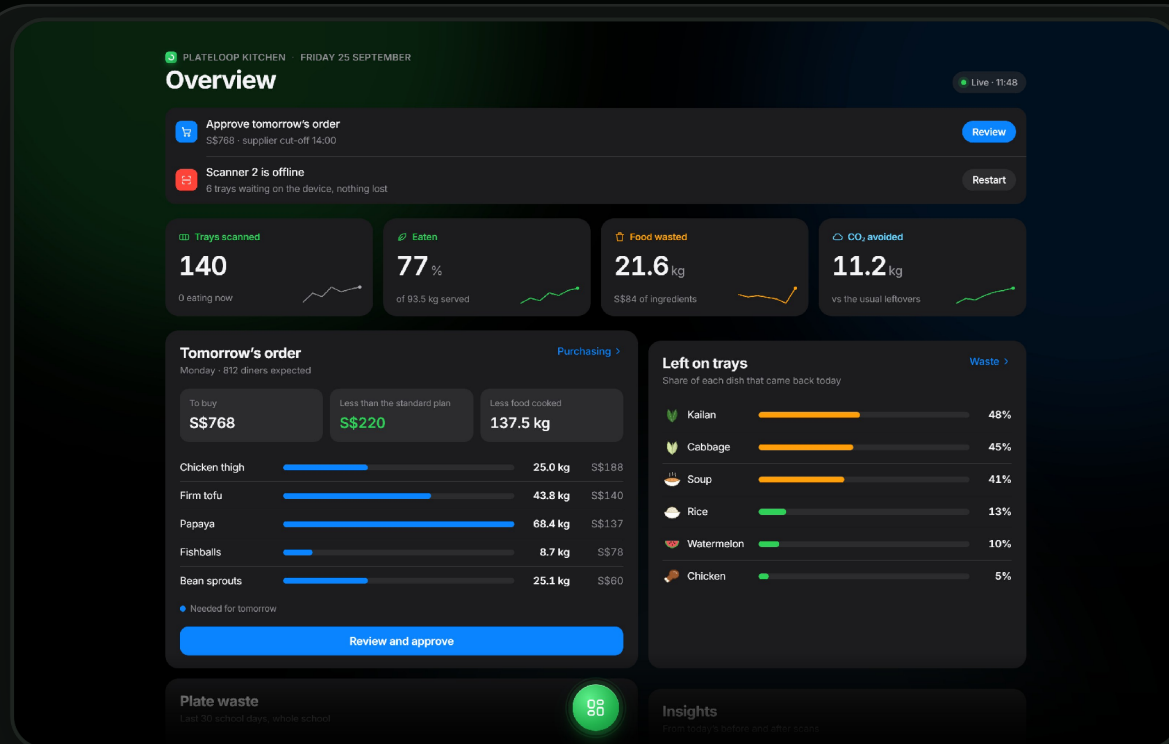
3D model, rendered in Blender



grams.

uncounted.

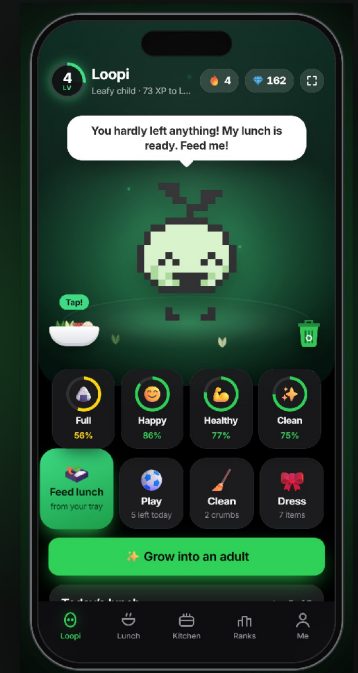
3 One scan, four apps



PlateLoop Kitchen

CANTEEN STAFF

- ✓ **Tomorrow's order** from real attendance, weather and events.
- ✓ **Waste by dish**, so the kailan nobody eats gets smaller.
- ✓ **Carbon ledger**: CO₂ avoided, ready for reporting.



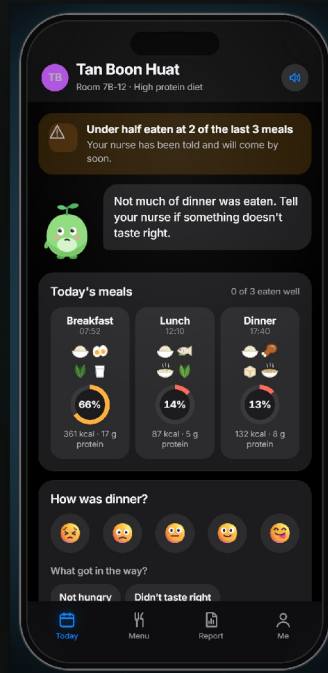
Loopi

STUDENTS

- ✓ A **virtual pet** fed by the lunch you
- ✓ **Food groups**, not calories.
- ✓ **Class races**, streaks and challenge

-35%

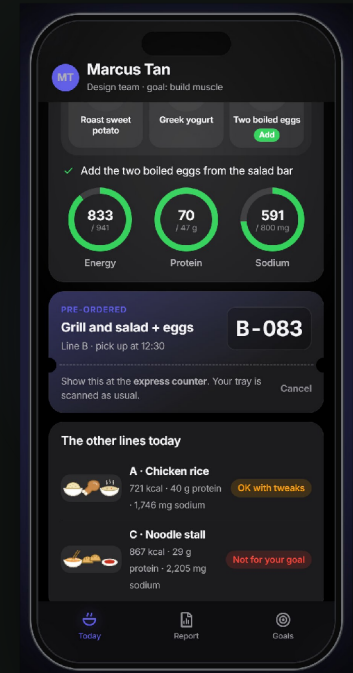
each built from research on the people



Loopi Care HOSPITALS

- ✓ Every tray measured, no food charts.
- ✓ Patients **choose tomorrow's meals**.
- ✓ Alerts reach the nurse **with the reason**.

29%



Loopi Work OFFICES

- ✓ A **daily pick** for your health goal.
- ✓ **Pre-order** and skip the queue.
- ✓ **Private**: teams, never names.

6 in 10



£\$28,500

ingredients saved a year in an 800-meal school (illustrative)

–35%

plate waste when classes weighed a
their waste (Swedish school study, 2

4 Impact

5.7t

less food thrown away

14.3t

CO₂e avoided

5.1x

return on t

Evidence: 86 catering sites that measured their waste cut it 36% in a year, with a 6:1 return (Champions 12.3, 2018). Our estim

29%

of inpatients were malnourished (Lim et al.,
Clinical Nutrition 2012)

6 in 10

Singaporeans eat out at least 4 times a
(HPB)

one 800-meal school, a year

1x

on the subscription

2.4 mo

to pay back a year



Try
8jiwo
plate

estimate assumes a smaller 30% cut, 25% wasted today, S\$2.50 a meal. To validate in a pilot.

es a week

Try it live

[gijwoo.github.io/
plateloop](https://gijwoo.github.io/plateloop)